AI Data Center
Power and Compute Co-Scheduling

HackSHSID 2026 Track 2 | Public-data study | 2020-06-01 energy day

Purpose and result

Flexible computing and batteries both move electricity demand across time. We coordinate non-preemptive GPU job starts and two storage layers to reduce operating cost while preserving admitted work and an explicit carbon policy.

All 26 jobs finish before their assumed deadlines. Joint green reduces modeled cost by 10.97% and peak import by 30.62%, while meeting a chosen 3% carbon-accounting reduction. Both batteries restore initial energy.

Metric	FIFO	Joint green
Cost proxy / CNY per day	230.85	205.54
Carbon accounting / kg CO2	97.67	94.74
Peak grid import / kW	42.55	29.52
Modeled GPU-hours	762.00	762.00

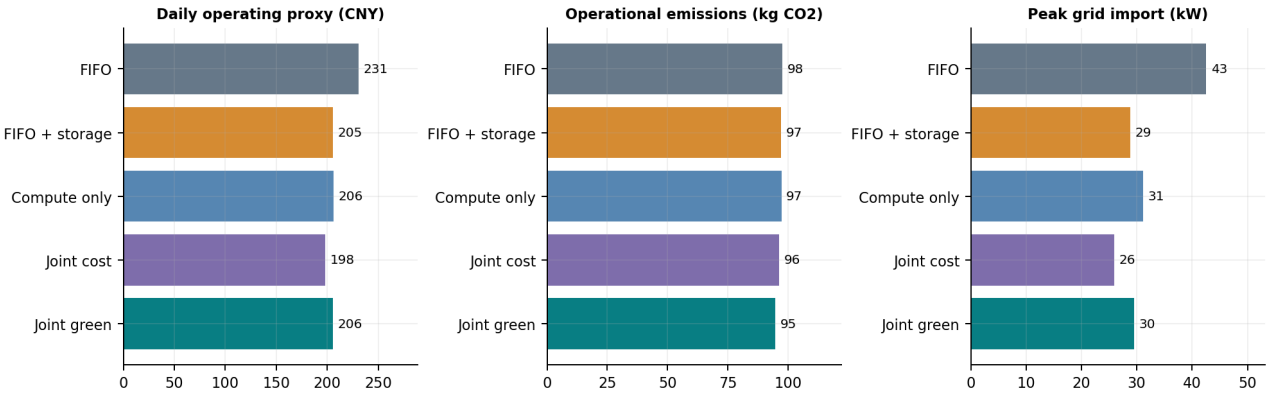


Figure 1. Same admitted workload; all five saved policies independently audited.

The benchmark combines observed Helios GPU jobs, OPSD British electricity/solar records and UK carbon estimates. It is a counterfactual composite, not a measured British facility: deadlines, power, assets, currency conversion, national-solar scaling and workload relocation remain assumptions.

SciPy/HiGHS solves a start-indexed mixed-integer linear program. The green objective gap is 0.000055%. Other public dates include unattainable targets; constructed stresses include negative cost savings. Neither failures nor work are discarded.

2 Data and study assumptions

Input	Source and transformation
GPU jobs	Helios Saturn / vcOIr: completed GPU jobs, 1,800-10,800 s, submitted before 18:00; first at most 32 by submission then source ID
Capacity/background	192 published VC GPUs; fixed background is each slot's maximum concurrency from same-VC completed GPU jobs shorter than 1,800 s
Electricity	OPSD GB_GBN_price_day_ahead, GBP/MWh x assumed 9 CNY/GBP / 1,000; negative prices retained
Solar	OPSD actual national solar MW x assumed 0.004 kW/MW; missing capacity factors are not filled
Carbon	UK national API actual field, estimated average g CO2/kWh / 1,000

The 26 source jobs retain IDs, CSV rows, GPU counts, submit/start/end times and execution seconds. Upward 15-minute rounding gives 762.00 modeled GPU-hours from 693.86 observed GPU-hours (9.82% padding). Deadlines allow up to four hours waiting after rounded duration, capped at day end. No dependency graph or inference label is invented.

Energy records align to UTC interval starts. Helios timestamps have no documented timezone; within-day offsets are rebased onto the modeled British day. Hourly/half-hourly observations are held constant, without interpolated noise. The completed-job subset is not a representative sample of all Helios work.

Assumption	Configured value
Base / active or reserved GPU / PUE	20 kW / 0.25 kW per GPU / 1.20
Facility / grid / successive ramp	150 kW / 120 kW / 50 kW per step
Renewable cost / daily peak proxy	0.06 CNY/kWh / 2 CNY per kW-day
LFP / sodium-ion capacity	40 kWh / 6 kWh

Helios is commit/archive-hash pinned. OPSD's database was recovered through a pinned archive mirror and exactly matches the publisher's SHA-256; the mirror is a transport route. Source excerpts, metadata, terms and hashes ship in data/raw/. Missing/corrupted required observations fail explicitly.

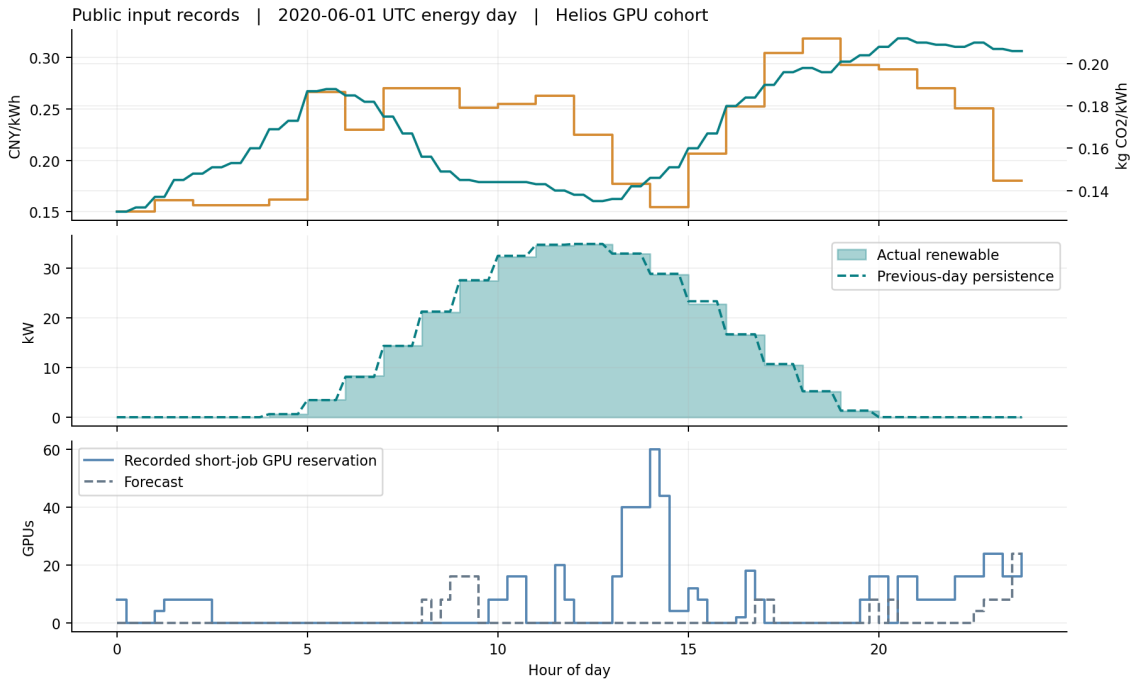


Figure 2. Public signals and derived inputs; solar/background forecasts use previous-day persistence.

3 Job scheduling and power balance

Symbol	Meaning and unit
t, s; delta t	Slot/start indices; slot duration 0.25 h
r, d, L, W; g, p	Release, exclusive deadline, duration, max wait in slots; GPUs, incremental kW
z, a; n, N	Binary start and derived activity; background GPUs and capacity
G, R, c, u; E	Grid, renewables, battery charge/discharge kW; stored energy kWh
kappa, pi, xi	PUE; converted wholesale CNY/kWh; average kg CO2/kWh

$$\mathcal{F}_j = \{s \in \mathbb{Z} : r_j \le s \le \min(d_j - L_j, r_j + W_j)\}, \qquad \sum_{s \in \mathcal{F}_j} z_{js} = 1. \tag{1}$$

$$a_{jt} = \sum_{\substack{s \in \mathcal{F}_j \\ s \le t < s + L_j}} z_{js}, \qquad \sum_j g_j a_{jt} + n_t \le N_t. \tag{2}$$

Exactly one start executes each admitted job for L consecutive slots; finishing at d is allowed, executing in slot d is not. The main pool is homogeneous GPU equivalents. Host memory/CPU and fabric placement remain outside this aggregate model.

$$\sum_{s \in \mathcal{F}_k} s z_{ks} \ge \sum_{s \in \mathcal{F}_j} (s + L_j) z_{js}, \qquad (j,k) \in \mathcal{E}_{\text{dep}}. \tag{3}$$

Parent completion precedes child start when a DAG is supplied. This capability is tested; the public cohort has no supplied edges.

$$P_t^{\text{IT}} = P_{\text{base}} + q_{\text{inf}} n_t + \sum_j p_j a_{jt}, \qquad P_t^{\text{fac}} = \kappa_t P_t^{\text{IT}}. \tag{4}$$

$$G_t + R_t + \sum_b u_{bt} = P_t^{\text{fac}} + \sum_b c_{bt}. \tag{5}$$

$$\begin{aligned} 0 \le G_t \le \bar{G}_t, \quad 0 \le R_t \le \bar{R}_t, \quad P_t^{\text{fac}} \le \bar{P}^{\text{fac}}, \\ |G_t - G_{t-1}| \le \Gamma \quad (t \ge 1). \end{aligned} \tag{6}$$

Base IT power and incremental job/background power are distinct. Renewables may be curtailed; exports are forbidden. Grid ramp applies between successive slot averages, with no previous-day boundary. Legacy inference_gpus/training_gpus fields denote background/scheduled occupancy, not source workload labels.

4 Storage, objective and solver

$$E_{b,t+1} = E_{bt} + \eta_b^c c_{bt} \Delta t - \frac{u_{bt} \Delta t}{\eta_b^d}. \tag{7}$$

$$E_b^{\min} + E_b^{\text{res}} \le E_{bt} \le E_b^{\max}, \qquad E_{b0} = E_{bT} = E_b^{\text{init}}. \tag{8}$$

$$\begin{aligned} &0 \le c_{bt} \le (\bar{c}_b - P_b^{\text{res}}) v_{bt}, \\ &0 \le u_{bt} \le (\bar{u}_b - P_b^{\text{res}})(1 - v_{bt}), \qquad v_{bt} \in \{0, 1\}. \end{aligned} \tag{9}$$

Asset	**Initial / limits kWh**	**Charge/discharge kW**	**Efficiencies**
LFP	20 / 4-36	10 / 10	0.96 / 0.96
Sodium-ion	3 / 0.6-5.4	6 / 6	0.94 / 0.94

Sodium-ion retains an extra 1.2 kWh and 2 kW in each direction. Both batteries restore initial energy. Per-battery exclusive modes permit different layers to operate in opposing directions; throughput wear penalizes cycling but does not forbid it under negative prices.

$$C = \Delta t \sum_t (\pi_t G_t + \pi_R R_t) + \lambda Q + \Delta t \sum_{b,t} k_b (c_{bt} + u_{bt}), \quad Q \ge G_t. \tag{10}$$

$$M = \Delta t \sum_t \xi_t G_t, \qquad \min C \quad \text{subject to} \quad M \le 0.97 M_{\text{FIFO}}. \tag{11}$$

C includes energy, renewable variable cost, a daily peak proxy and AC-throughput wear (LFP 0.035; sodium-ion 0.055 CNY/kWh). M uses average grid-intensity estimates. Charging is counted at import; discharge gets no extra carbon credit. Modeled renewable operational intensity is zero; lifecycle emissions and battery capex are excluded.

HiGHS uses mixed-integer branch-and-bound/cut methods through SciPy milp. Status 0 means requested tolerance reached, 1 a time/resource limit, 2 infeasibility. An LP relaxation supplies a lower bound but fractional jobs are not executable. Stricter-budget incumbents may improve looser problems; lower bounds remain target-specific.

$$L \le C^* \le U, \qquad \text{gap} = \frac{U - L}{|U|}. \tag{12}$$

Green U = CNY 205.535943792, L = CNY 205.535831654, gap = 0.000055%. The model has 1404 variables, 633 binaries and 1082 constraints. The independent audit reconstructs physical/accounting outcomes without reading the MILP matrix.

5 Dispatch and mechanism comparisons

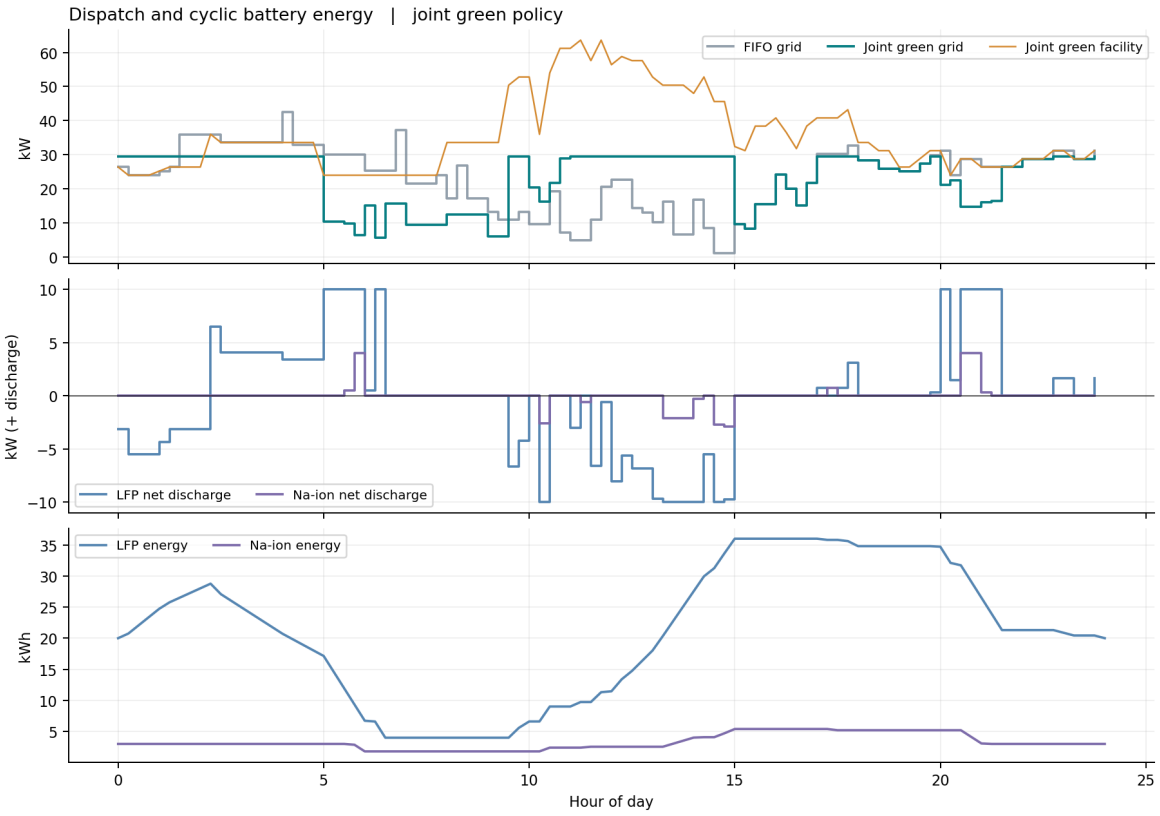


Figure 3. Recommended dispatch, battery net powers and all 97 energy states.

Facility demand is 855.60 kWh; grid import 563.46 kWh; renewable use 295.91 kWh. Conversion losses add 3.78 kWh. Cyclic terminal energy prevents apparent savings from emptying the initial stock.

Same 3% carbon policy	**Cost CNY**	**Gap / status**
FIFO + storage	Infeasible	Status 2
Compute only	Infeasible	Status 2
Joint, LFP only	206.47	0.092608%
Joint, both layers	205.54	0.000055%

Cost-only joint scheduling costs CNY 198.08; the selected green requirement adds CNY 7.45. Compute/storage benefits cannot be added as independent savings because each changes the other's operating opportunities.

The LFP-only comparison isolates the second layer's modeled economic increment, not fast-response capability or investment return. Sampled cost/carbon budgets expose trade-offs but do not establish a complete exact frontier. Job CSVs and the offline explorer retain every source-ID schedule.

6 Public dates and constructed stresses

Public date	FIFO service	Green policy	Cost saved
2020-06-02	Feasible	Feasible	17.28%
2020-06-03	Feasible	Infeasible	N/A
2020-06-04	Feasible	Feasible	0.52%
2020-06-05	Feasible	Feasible	13.02%
2020-06-08	Feasible	Infeasible	N/A

June 3 and June 8 have feasible cost-only service, but valid minimum-carbon lower bounds exclude the extra 3% target. The same fixed selection rule applies to all dates. These related observations do not form a facility-wide statistical guarantee.

Constructed scenario	FIFO	Green	Cost saved
cloudy	Yes	Yes	-6.75%
no renewables	Yes	No	N/A
flat price	Yes	Yes	9.87%
price carbon conflict	Yes	Yes	13.47%
hot day	Yes	Yes	8.30%
Background GPU surge	Yes	No	N/A
grid derating	Yes	Yes	10.94%
zero flexibility	Yes	No	N/A
storage expensive	Yes	Yes	-2.34%
no demand charge	Yes	Yes	5.22%
high workload	Yes	Yes	12.63%
impossible grid	No	No	N/A

These are perturbations, not historical incidents: cloudy retains 35% solar; background surge adds 48 GPUs; grid derating halves its cap; hot day sets PUE 1.40. Other cases vary waiting, prices, charges, wear and workload. Negative savings and infeasible targets are retained.

Impossible grid sets both grid and renewable supply to zero. Cyclic batteries cannot create net daily energy; a positive supply-deficit diagnostic proves failure. Passing this necessary energy condition alone does not prove time/resource feasibility. N/A denotes no feasible same-scenario saving comparison.

Independent minimum-carbon runs also separate unattainable environmental targets from workload failures in constructed cases. All five public dates and twelve stresses remain in results/main/.

7 Online replay, validation and references

Online policy	Cost CNY	Carbon kg	Peak kW
Default reserve 4	228.41	98.27	41.20
Selected reserve 0	226.94	98.11	40.64

Both replays complete 96 decisions. Past actions/launched starts are locked and future releases hidden. Future solar/background use previous-day persistence; current-slot aggregates are optimistically assumed known at the boundary. Archived carbon forecasts have unknown vintage and do not affect this cost-only policy.

Reserve 0 is selected from 0/2/4/8 on June 2-3 by failures then cost; June 4-5 are separate evaluation. Both default and selected margins survive the declared shock (solar 15% in slots 40-55, +48 reserved GPUs in 44-51). Neither replay guarantees the offline carbon cap or a global online optimum.

Validation passes 74 unit tests and 55 numerical audits. Sixty brute-force random instances are subcases of three tests. Regression cases reject duplicate/unknown batteries, invalid time steps and nonfinite arithmetic. CLI, browser and PDF QA are recorded separately.

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Limits: assumed deadlines/power/storage, homogeneous GPUs, average carbon and daily peak proxies. No verified request latency, CPU/RAM/topology, transient response, monthly utility bill, TCO/ROI or causal grid-emissions reduction.

References

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[2] Open Power System Data, Time series 2020-10-06. Primary ENTSO-E records; source-specific terms retained.

[3] UK Carbon Intensity API. National estimates and archived forecasts. CC BY 4.0.

[4] SciPy milp: HiGHS interface, statuses and certificates.

[5] Radovanovic et al. 2021. Carbon-Aware Computing for Datacenters.

[6] HiTHIUM storage announcement. Motivation; modeled asset ratings remain assumptions.

[7] Supplied HackSHSID Track 2 statement, project brief and technical specification. [8] IMMC 2025 Problem E, When Watts Meet Bits. Source identities/hashes: data/source_documents.json. Public-data URLs/hashes/terms: data/raw/manifest.json.